higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# NATIONAL CERTIFICATE (VOCATIONAL)

## PROCESS CHEMISTRY
### NQF LEVEL 2

(6050422)

**1 March 2019 (X-Paper)**
**09:00–12:00**

**This question paper consists of 5 pages.**

NCV

**TIME: 3 HOURS**
**MARKS: 100**

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**INSTRUCTIONS AND INFORMATION**

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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NC (V)

## QUESTION 1

Give ONE word/term for each of the following descriptions. Write only the word/term next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1 A personal protective device that protects the ears.

1.2 An industrial gas that is used as a fire extinguisher.

1.3 Recycled glass that is added to raw materials during the production of glass.

1.4 An allotrope of oxygen that is used as a disinfectant by the water purification industry.

1.5 A company formed by the South African government in 1950 to produce petroleum from coal.

(5 × 1) **[5]**

## QUESTION 2

2.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.1.1–2.1.5) in the ANSWER BOOK.

2.1.1 Soda feldspar contains 68.5% silica.

2.1.2 Oxygen is used as a rocket fuel.

2.1.3 Water for flushing toilets is mixed with human waste, soap and saliva.

2.1.4 Herbicides are chemicals that are used to control the spread of unwanted plants.

2.1.5 The National Water Act sets specifications for water for domestic and industrial purposes.

(5 × 1) (5)

2.2 Write the full names of the following chemical industrial companies:

2.1.1 NECSA

2.1.2 PPC

2.1.3 Sappi

2.1.4 Eskom

2.1.5 TRUCO

(5 × 1) (5)
**[10]**

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## QUESTION 3

3.1 Name FOUR examples of base metals. (4)

3.2 Name THREE paper-producing companies in South Africa. (3)

3.3 Name THREE main components of natural gas. (3)

3.4 Briefly discuss how coal is recovered through underground mining. (3)

**[13]**

## QUESTION 4

4.1 Water is important for both domestic and industrial purpose.

4.1.1 Name FOUR physical properties of water. (4)

4.1.2 Explain how the COD level of water is determined. (2)

4.2 Write the FIVE steps of municipal waste water treatment in sequence. (5)

4.3 Describe water hardness. (4)

4.4 Write brief, clarifying notes on chemical dosing in portable water treatment. (5)

4.5 Give FIVE types of chemicals that are added to the chemical dosing process. (5)

4.6 How would you add chlorine gas to disinfect drinking water? (2)

**[27]**

## QUESTION 5

5.1 Name THREE uses of chlorine. (3)

5.2 Name FIVE advantages of the industrial gas industry. (5)

5.3 Write a short description of the following:

5.3.1 Distillation

5.3.2 Greenhouse effect

(2 × 2) (4)

5.4 Discuss the working principle of refrigeration. (6)

**[18]**

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## QUESTION 6

6.1 Draw a process diagram of glass production. (12)

6.2 Name THREE types of potash feldspar. (3)

6.3 Give THREE methods of heat transfer that occur in a glass furnace. (3)

6.4 Name THREE characteristics of hazardous chemicals. (3)

6.5 The purpose of applying environmentally friendly disposal methods is to prevent harm to the water, land and air.

6.5.1 Name THREE methods that are regarded as improper disposal of hazardous chemicals.

6.5.2 Name THREE correct disposal methods of hazardous chemical containers.

(2 × 3) (6)
**[27]**

**TOTAL: 100**

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